

The Political Economy of Fragmented Housing Regulation

By-Right Envelope, Discretionary Friction, and the Minutes Record across a Single Labor Market

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Abstract

The “abundance” diagnosis holds that housing markets in productive U.S. metros are out of equilibrium because land-use regulation throttles supply [Klein and Thompson, 2025, Hsieh and Moretti, 2019, Glaeser et al., 2005a]. What that literature largely takes as exogenous—the stringency of regulation—is the equilibrium output of a *market for regulation* run through local bodies. This note builds a deliberately small model with two changes relative to the supply-elasticity tradition. First, regulation operates on *two margins*: a *by-right envelope* set by the zoning map (what may be built ministerially, with certainty and no hearing) and a *discretionary friction* applied to everything that exceeds it (conditional-use authorization, discretionary review, conditions, and delay). The empirical fact that motivates this split is that approvals, *conditional on reaching a vote*, are nearly universal: restriction does not operate through denial at the hearing but through what must be endured to reach it. Second, the metro is *fragmented*: one integrated labor market and a handful of housing submarkets are regulated by ~ 100 jurisdictions whose boundaries align with neither, so each jurisdiction’s choice of both margins is the solution to a game played against its neighbors [Brueckner, 1998, Fischel, 2001]. Every object maps to a field in an item-level dataset built from planning-commission minutes (1998–present) under a single canonical schema. The point is measurability: a model whose primitives are the things the minutes record, and—crucially—a model that locates restriction where the record shows it actually lives.

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1 Introduction and relation to the literature

A large body of work establishes that the post-1970 divergence between house prices and construction costs in high-demand metros is a supply phenomenon driven by land-use regulation [Glaeser et al., 2005a,b, Glaeser and Ward, 2009, Quigley and Rosenthal, 2005], and that the resulting spatial misallocation is macroeconomically large [Hsieh and Moretti, 2019]. Measurement has advanced from survey indices [Gyourko et al., 2008, 2021, Gyourko and Molloy, 2015] to LLM-based generative regulatory measurement [Bartik et al., 2024]. Yet both approaches measure regulatory *outcomes*; neither observes the *process* that produces them.

The political-economy ingredients are well known in isolation. Fischel [2004] formalizes the “homevoter”; Ortalo-Magné and Prat [2014] study homeownership versus affordability in a voting model; Jones et al. [2022] document how partisanship shapes housing policy; Weiss [1988] traces the real-estate industry’s hand in San Francisco’s original zoning regime.

What is new here. Three features the existing work abstracts away are exactly the features the administrative record makes visible. (i) Regulation is *two instruments, not one level*: the map sets a by-right envelope, and a separate discretionary process taxes whatever exceeds it. (ii) That discretionary tax operates through *process friction*—the requirement of a hearing, the mobilization of opposition, conditions, and delay—rather than through denial, which is rare. (iii) The metro is *fragmented*, so both instruments are chosen jurisdiction-by-jurisdiction in a game coupled through shared prices and a shared labor market, making spillovers and strategic interaction first-class objects [Brueckner, 1998, 2003]. The contribution is a tractable equilibrium object with all three features, stated in terms a researcher can measure item-by-item across many jurisdictions.

2 Environment

One productive metro—the Bay Area—comprises J jurisdictions $j = 1, \dots, J$ and a competitive “rest of country” delivering reservation utility $\bar{u}$. The defining feature is a mismatch of boundaries: *one labor market* (all jurisdictions draw on a common worker pool; a worker earns the regional wage $w_m = w(N)$, $w' > 0$, $N \equiv \sum_j N_j$, capturing Hsieh–Moretti agglomeration rents), *a few housing submarkets* ($s(j) \in \{1, \dots, S\}$, $S \ll J$, within which housing is a close substitute), and *many regulators* (each j sets its own regulation).

Preferences and prices. A household of type θ in j has utility $U = c + \alpha \ln h - \gamma g(D_j)$, $g' > 0$, with $-\gamma g(D_j)$ the local density disamenity (so low density is a normal good and high- θ residents are the natural demanders of restriction), and income $y(\theta) = w_m \theta$ regardless of residence—the labor-market coupling. Within submarket s , clearing gives the price index

$$p_s = \frac{\alpha N_s}{H_s}, \quad N_s = \sum_{j:s(j)=s} N_j, \quad H_s = \sum_{j:s(j)=s} H_j, \quad (1)$$

so a unit built in any jurisdiction of s lowers the price faced by incumbents in every jurisdiction of s ; density $D_j = N_j/\bar{L}_j$ is local. The cost of new housing is local; the benefit (lower submarket price, more regional employment hence higher w_m) is shared.

3 Regulation on two margins

This section replaces the single “approval rate” of earlier drafts with the structure the record actually exhibits. The motivating fact, visible in the minutes: among items that reach a commission vote, approval (or its discretionary-review equivalent, declining to intervene) is nearly universal. Denial is not the instrument. Restriction operates *before and around* the vote, on two margins.

The by-right envelope. The zoning map assigns each parcel in j a use district and a height-and-bulk district, which together define a *by-right envelope* $\bar{z}_j$: the scale and use a project may obtain *ministerially*—built with certainty, no hearing, no discretionary cost. A project with proposed scale/use $z \leq \bar{z}_j$ clears by right and never appears on the commission calendar. This is why the commission’s *conditional* approval rate is uninformative about stringency: the by-right projects, the bulk of conforming construction, are invisible to the hearing record by construction.

The discretionary tax. A project that requires more than the envelope allows—a use not principally permitted, a scale above the limit, a parcel in a special district—must enter discretionary review (conditional-use authorization, variance, mandatory or requested discretionary review). It is built only after enduring a *discretionary tax* τ_j whose components are exactly what the minutes record. Each component below names the schema field(s) that identify it and how it is obtained—directly labeled, derived post-hoc, or (for two components) deferred pending a second source:

1. **Process exposure**—the requirement of a hearing at all, i.e. the share of activity the envelope forces off the by-right path. *Labeled:* the `request_type` enum (conditional_use, discretionary_review, variance, ...) and `special_use_district`.
2. **Mobilized opposition**—the disamenity $g(D_j)$ made politically operative. *Labeled (modern era $\approx 2015+$):* `oppose_count/support_count/neutral_count` (stance markers + / - / =) and the grounds in `speaker_statements`.
3. **Disposition and conditioning**—whether the commission approved, declined to intervene, conditioned, or denied. *Labeled:* the `action` enum, which for discretionary-review items uses the dedicated values `did_not_take_dr / took_dr / took_dr_and_approved`; the *margin vote* is *derived* in coercion from $|\text{yes}| - |\text{noes}|$. The conditioning *content* (`modifications` free text) is captured where present but is sparse; treat conditioning as an indicator (`action=approved_with_conditions / approved_as_modified`) rather than a measured intensity.
4. **Delay**—continuances, which raise carrying cost and induce attrition. *Derived (post-hoc):* `action=continued` plus a case-level rollup—hearings per case and first-to-final-disposition span—built by joining item records on `case_number` across meetings. The item-level schema does not express this directly; it is a downstream table, not a labeled field.
5. **Denial**—rare, but present. *Labeled:* `action=disapproved` or `took_dr` against the project.

Two finer measures of the tax—the project’s scale *above* the by-right envelope, and the affordable/exaction concessions extracted—are *not* captured in this build: proposed stories/height are deferred to a derived pass over `project_descr`, and inclusionary terms live in staff reports outside the minutes corpus. The model therefore identifies τ_j from process exposure, opposition, disposition/conditioning, delay, and denial, and treats the scale-above-envelope and exaction margins as extensions contingent on those additional sources.

Definition 1 (Effective regulation as a composite of two margins). *Effective stringency in j is $r_j = g(\bar{z}_j, \tau_j)$, increasing in the tightness of the by-right envelope (smaller $\bar{z}_j$ pushes more activity into the discretionary channel) and in the discretionary tax τ_j (heavier process exposure, opposition, conditioning, and delay). Crucially, r_j can rise with no change in the conditional approval rate: a jurisdiction restricts by shrinking $\bar{z}_j$ or raising τ_j , not by denying more of what reaches a vote.*

The developer’s problem. A candidate project of latent scale/use z chooses how far to push above the envelope. Surplus from scale z is $\Pi(z)$, increasing and concave; the discretionary cost of exceeding the envelope is $\tau_j \cdot \max\{z - \bar{z}_j, 0\}$. The developer builds $z_j^* = \arg \max_z \Pi(z) - \tau_j \max\{z - \bar{z}_j, 0\}$, files at z_j^* if the net surplus is positive, and otherwise withdraws (or never files). Three empirical signatures follow: (i) bunching of projects at the envelope $\bar{z}_j$ (testable once proposed scale is recovered from `project_descr`); (ii) near-universal approval *conditional on filing above the envelope*, because projects that would be denied are deterred or shaved first—directly visible in the `action` distribution; (iii) restriction showing up as withdrawals, continuances, conditions, and a thin above-envelope calendar, not as denials—visible in `action` and the case-level continuance rollup.

Definition 2 (New supply). *New construction in j aggregates by-right building (projects with $z \leq \bar{z}_j$, built with certainty) and surviving above-envelope projects: $\Delta H_j = \kappa_j^{BR} + \int \mathbf{1}\{\Pi(z_j^*) - \tau_j(z_j^* - \bar{z}_j) \geq 0\} z_j^* dF(z)$. The discretionary record observes only the second term; the first is recovered from the map and permit data.*

4 The political mechanism: homevoters choose $(\bar{z}_j, \tau_j)$

Following the homevoter logic [Fischel, 2004], jurisdiction j ’s authorities are vote-maximizers answering to incumbent residents, a share λ_j of whom are homeowners holding local housing as their principal asset; ϕ_j is the political weight on incumbents. The two margins map naturally to two bodies: the *city council* sets the map $\bar{z}_j$ (legislative, infrequent), and the *planning commission* sets the discretionary tax τ_j through its revealed disposition behavior (case-by-case, high-frequency). This resolves the “unitary authority” shortcut of earlier drafts: the map is a slow state variable, the discretionary tax a fast one.

An incumbent homeowner’s payoff from her jurisdiction’s regulation, taking neighbors’ policies as given, is

$$W_j^{\text{own}} = \underbrace{p_{s(j)}(\bar{z}_{-j}, \tau_{-j}; \bar{z}_j, \tau_j)}_{\text{submarket asset value}} - \underbrace{\gamma g(D_j(\bar{z}_j, \tau_j))}_{\text{local density disamenity}}, \quad (2)$$

where both $\bar{z}_j$ and τ_j raise the asset value (by restricting local supply) and reduce the disamenity, while only *regional* supply matters for the price an incumbent’s asset commands. Renters/entrants (weight $1 - \phi_j \lambda_j$) prefer abundant regional supply. The authority chooses $(\bar{z}_j, \tau_j)$ to maximize $\Omega_j = \phi_j \lambda_j W_j^{\text{own}} + (1 - \phi_j \lambda_j) W_j^{\text{ent}}$.

The fragmentation wedge. As in the single-margin model, the effect of j ’s own supply on its submarket price is attenuated by submarket size: $\partial p_{s(j)} / \partial(\text{supply}_j) = -(p_s / H_s)$. A jurisdiction small within its submarket bears the full local disamenity of any approval but captures little of the price relief (it leaks to co-submarket neighbors), so it restricts on *both* margins.

Proposition 1 (Substitution between the two margins under preemption). *Suppose state preemption (e.g. ministerial-approval mandates, by-right upzoning, a builder’s remedy) raises the marginal*

cost to j of a tight envelope—forcing $\bar{z}_j$ up or converting discretionary categories to ministerial ones—while leaving the marginal cost of discretionary friction τ_j unchanged. Then a jurisdiction whose homevoter bloc still demands restriction substitutes from the envelope to the tax: $\bar{z}_j$ rises (the map loosens, as the mandate intends) while τ_j rises in compensation (more items pushed to conditions, more opposition entertained, more continuances). Effective stringency $r_j = g(\bar{z}_j, \tau_j)$ falls by less than the change in $\bar{z}_j$ alone would imply, and the conditional approval rate does not move. A scalar index, or an approval-rate measure, registers the reform as working; the decomposed record reveals the offset.

This is the central testable mechanism and the reason both data margins are required. It is invisible to any one-dimensional measure of stringency.

Proposition 2 (Fragmentation intensifies restriction on both margins). *Holding $(\phi_j \lambda_j, \gamma, \bar{L}_j)$ fixed, a jurisdiction’s restriction on each margin is increasing in the number of jurisdictions sharing its housing submarket. As j becomes atomistic within s , the price-relief motive vanishes, the local-disamenity motive dominates, and both $\bar{z}_j$ tightens and τ_j rises toward their restrictive corners.*

5 Strategic interaction and regional equilibrium

Because $p_{s(j)}$ depends on neighbors’ regulation, (2) defines a game with reaction functions on each margin.

Proposition 3 (Regulation is strategically substitutable within a submarket). *For co-submarket jurisdictions, $\partial R_j / \partial(\text{loosening}_k) < 0$: if a neighbor loosens (raises $\bar{z}_k$ or lowers τ_k), the submarket price falls, raising a homevoter-pivotal j ’s incentive to restrict. Each jurisdiction free-rides on others’ supply; the Nash equilibrium has less regional construction than the submarket-coordinated optimum.*

Proposition 4 (Regional over-restriction and demand-shock amplification). *A metro planner internalizing entrants’ access to the agglomeration rent and the cross-jurisdiction price externality chooses looser regulation on both margins for every j where the homevoter bloc is pivotal; the wedge grows in $\phi_j \lambda_j$, in the agglomeration rent, and in submarket fragmentation. A positive regional demand shock raises every p_s , raising the asset-value return to restriction, so each jurisdiction tightens; strategic substitutability (Prop. 3) propagates the tightening across co-submarket neighbors. The regional price response exceeds both the fixed-regulation and the single-jurisdiction response.*

6 Extensions

Credit / CRA. An easing raises the demand shifter $\alpha \rightarrow \alpha(1 + \mu)$ [Favara and Imbs, 2015, Di Maggio and Kermani, 2017], raising p_s and (Prop. 4) tightening regulation, so part of a credit boom shows up as tighter discretion rather than more building. The CRA’s tract-level eligibility generates within-metro, within-submarket variation in μ [Saadi, 2020, Brevoort, 2024]; the minutes locate each project (address $\rightarrow$ tract $\rightarrow$ jurisdiction $\rightarrow$ submarket), so the relevant object is the interaction of tract credit shocks with the local discretionary response.

Dynamics: ratchet and contagion. Let the homeowner share evolve with prices, $\lambda_{j,t+1} = \lambda_{j,t} + \eta(p_{s(j),t} - \bar{p})$. With submarket price linkage and strategic substitutability, a shock that tips one jurisdiction into a restrictive basin raises the submarket price, entrenching homeowners and tipping co-submarket neighbors: the restrictive regime is sticky within and *contagious across* jurisdictions. This motivates the long panel (1998–present, $\geq$ one price cycle) across jurisdictions.

7 Mapping the model to the minutes data

The dataset is built item-by-item from planning-commission minutes under a single canonical schema (35 typed fields; one record per agenda item), with machine pre-fill corrected by a human in a two-pane labeling app and scored field-by-field on a held-out split. Table 1 maps each model object to the schema field(s) that identify it, with the empirical caveats that the records impose.

Toward identification. Reduced-form tests, ordered by how cleanly the current data supports them:

- **Staff-commission override (Prop. 4):** the cleanest, because it is a *within-item* comparison—regress an override-toward-restriction indicator (`preliminary_recommendation` more permissive than `action`) on the project tract’s homeownership share and on `oppose_count`. The override indicator is *derived* by normalizing `preliminary_recommendation` onto the `action` enum—a post-processing step, not a relabeling cost.
- **Margin substitution under preemption (Prop. 1):** difference-in-differences around SB-9/35/423 / builder’s-remedy exposure, with the *envelope* (map) and the *discretionary tax* (minutes) as separate outcomes; predict $\bar{z}$ loosens while τ rises and the conditional approval rate is flat.
- **Strategic interaction (Prop. 3):** spatial-lag reaction function of j ’s discretionary tax on co-submarket neighbors’ tax, instrumented by neighbors’ exogenous characteristics [Brueckner, 1998, 2003]. Needs the multi-jurisdiction build.
- **Feedback / amplification (Prop. 4):** regress discretionary friction (opposition, continuances, conditioning) on lagged submarket price growth, instrumenting price with regional demand or rate shocks.
- **Mobilization $\rightarrow$ disamenity:** code `speaker_statements` into grounds (shadow, traffic, character, precedent) to test whether opposition intensity tracks the density terms the model assigns to $g(D_j)$.

8 What the build captures, derives, and defers

The schema (now 36 fields) and its coercion settle which model objects are labeled directly, which are derived downstream, and which are deferred to a second source. This division determines which tests run on this build.

- **Labeled directly.** Disposition (`action`, including the dedicated DR outcomes `did_not_take_dr/took_dr/took_dr`), process exposure (`request_type`), opposition (`oppose_/support_/neutral_count`, modern era), the roll-call (`ayes/noes/recused`), zoning districts, the assigned planner (`staff_planner`), and the spatial/time keys. These support the override, fragmentation, strategic-interaction, and amplification tests directly.
- **Derived, not labeled.** The vote margin `vote` is computed in coercion from `|ayes| - |noes|` (never overwriting a stated tally). The staff-commission override indicator is a post-processing comparison of `preliminary_recommendation` (normalized onto the `action` enum) with `action`. The delay margin—hearings per case and first-to-final-disposition span—is a case-level rollup joining item records on `case_number`. None consume labeling surface.

- **Deferred to additional sources or passes.** The project’s scale *above* the by-right envelope (proposed stories/height) is recoverable from `project_descr` but is left to a validated derived pass, because construction blocks carry multiple story figures (existing, addition, total) that a naive extractor confuses; until that pass is validated, the scale-bunching signature is untested. The discretionary *trigger* (which envelope margin binds) requires a code-section→category lookup and is deferred to a post-hoc NLP pass emitting a list. Affordable/inclusionary terms (the exaction margin) live in staff reports outside the minutes and are deferred pending that corpus. Final disposition after appeal lives in the Board of Supervisors / Board of Appeals series and is out of scope for a minutes-only build.

The practical consequence for the model: the discretionary tax τ_j is identified in this build from process exposure, opposition, disposition/conditioning, derived delay, and denial. The scale-above-envelope and exaction refinements (§3) are extensions that switch on when the deferred sources are added—they sharpen τ_j but are not required for the central two-margin substitution test (Prop. 1), which needs only the map (envelope) and the discretionary disposition record (tax).

9 Conclusion

Two moves—splitting regulation into a by-right envelope and a discretionary tax that operates through process friction rather than denial, and embedding many such jurisdictions in a game coupled through shared prices and a shared labor market—deliver a model whose restriction lives where the minutes show it lives, whose central prediction (substitution between the two margins under preemption, Prop. 1) requires exactly the two data dimensions the project builds, and whose strategic and dynamic results (Props. 3–4) motivate the multi-jurisdiction, long-panel extension. The near-universal conditional approval rate, far from a nuisance, is the equilibrium fingerprint of a regime that restricts before and around the vote rather than at it.

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Model object	Symbol	Schema field(s) and caveat
By-right envelope	$\bar{z}_j$	<code>type_district</code> , <code>height_and_bulk_district</code> , <code>special_use_district</code> (the map; the envelope is read off zoning, <i>not</i> the minutes) + permit data for by-right builds
Process exposure	component of τ_j	<code>request_type</code> composition (share forced into <code>conditional_use</code> / <code>discretionary_review</code> / <code>variance</code>)
Mobilized opposition	$g(D_j)$ operative	<code>oppose_count</code> , <code>support_count</code> , <code>neutral_count</code> (+/ - / =, <i>modern minutes only, $\approx 2015+$</i>); grounds in <code>speaker_statements</code>
Disposition & conditioning	component of τ_j	<code>action</code> enum (incl. <code>did_not_take_dr/took_dr/took_dr_and_approved</code> for DR; <code>approved_with_conditions/approved_as_modified</code> for conditioning); <code>vote</code> <i>derived</i> from <code> ayes - noes </code> . Conditioning treated as indicator— <code>modifications</code> text is sparse
Delay	component of τ_j	<code>action=continued</code> + case-level rollup (hearings per case, first-to-final span) <i>derived</i> by joining on <code>case_number</code> ; a downstream table, not a labeled field
Denial (rare)	component of τ_j	<code>action=disapproved</code> or <code>took_dr</code> against the project
Staff-commission wedge	planner vs. political	<code>preliminary_recommendation</code> vs. <code>action</code> : override rate, <i>derived</i> post-hoc by normalizing the staff recommendation onto the <code>action</code> enum
Planner	fixed effect	<code>staff_planner</code> (initial + surname from block header; near-universal, machine pre-filled)
Homevoter weight	$\phi_j \lambda_j$	<code>ayes/noes/recused</code> ; member-level approval propensity; board composition; tract homeownership (external)
Density / scale	z, D_j	<code>units_proposed</code> , <code>units_demolished</code> , <code>parking_spaces</code> (<i>reliable modern era; proposed stories/height deferred to a derived <code>project_descr</code> pass</i>)
Submarket / spatial keys	$s(j), W$	<code>project_address</code> →tract; <code>jurisdiction</code> ; <code>supervisory_district</code> ($\approx 2015+$); spatial-weights matrix over jurisdictions (external)
Price	p_s	External tract/ZIP indices merged on geocoded <code>project_address</code> , aggregated to submarket
Panel time index	t	<code>meeting_date</code>

Table 1: Model objects and their schema counterparts. Caveats note whether a field is labeled directly, *derived* downstream, or deferred; see §8.